

Qunzhong WANG

Department of Information Engineering
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Education

The Chinese University of Hong Kong

Double Major in Mathematics and Information Engineering

Overall GPA: 3.927/4.000

Hong Kong, China

Sep 2023 - Jul 2027 (expect)

University of Washington

Exchange Program in Computer Science and Engineering

Overall GPA: 4.000/4.000

Seattle, United States

Sep 2025 - Dec 2025

Selected Courses: Machine Learning, Deep Learning, Computer Architecture, Operating System, Distributed Systems, Database System, Real Analysis, Complex Analysis, Probability Theory, Stochastic Analysis

Research Interests

- **Principles of AI Systems backed by Math:** Understanding the mathematical principles behind model representation capacity, training dynamics, and generalization. Leveraging these principles to design better and more scalable architectures, optimizers, training methods.
- **Reinforcement Learning on Large Models:** Aligning Large Language Models (LLMs), Vision-Language Models (VLMs), and their derivative Agents with specific human preferences and demands, with techniques like Reinforcement Learning from Human Feedback (RLHF) and Reinforcement Learning with Verifiable Reward (RLVR).

Academic Experience

- **Department of Computer Science, Princeton University** Princeton, United States
Research Assistant, Advised by Prof. Zhuang Liu Sep 2025 - Present
 - **Research Focus:** Reasoning properties of VLM under supervised fine tuning & reinforcement learning
- **The Alan Turing Institute, The United Kingdom** Edinburgh, United Kingdom
Research Assistant, Advised by Prof. Sotirios Sabanis Jun 2025 - Aug 2025
 - **Research Focus:** Convergence analysis of stochastic algorithms in optimization algorithms.
- **Database Research Group, The Chinese University of Hong Kong** Hong Kong, China
Research Assistant, Advised by Prof. Hong Cheng May 2024 - Sep 2024
 - **Research Focus:** Provided key theoretical proofs for a transfer learning approach, Prompting, in Graph Neural Networks.

Industry Experience

- **Kling AI technology Department, Kuaishou Technology** Shen Zhen, China
Internship, Worked closely with fellows from MMLab, CUHK. Oct 2024 - Sep 2025
 - **Research Focus:** Post-Training of Video Generation Models and Adapting Vision-Language Models as Reward Models

Selected Publications¹

1. **Vero: Open Thinking Recipes for Visual Reinforcement Learning**
Gabriel Sarch[†], Linrong Cai[†], [Qunzhong Wang](#), Haoyang Wu, Danqi Chen, Zhuang Liu[‡]
Under review [Paper] [arXiv] [Code]
2. **VR-Thinker: Boosting Video Reward Models through Thinking-with-Image Reasoning**
[Qunzhong Wang*](#), Jie Liu*, Jiajun Liang[†], Yilei Jiang, Yuanxing Zhang, Yaozhi Zheng, Xintao Wang, Xiangyu Yue, Jiaheng Liu[‡]
Under review [Paper] [arXiv] [Code]
3. **Does Graph Prompt Work? A Data Operation Perspective with Theoretical Analysis**
[Qunzhong Wang*](#), Xiangguo Sun*, Hong Cheng[‡].
International Conference on Machine Learning (ICML), 2025 [Paper] [arXiv] [Code]

¹* for co-first author, [†] for project leader, [‡] for corresponding author.

Honors & Awards

- Talent Development Scholarship (HK\$10,000 awarded by HK Government) 2025
- Professor Charles K. Kao Research Exchange Scholarships (HK\$50,000 awarded by CUHK) 2025
- Dean’s List, CUHK 2024, 2025
- Meritorious Winner, International Mathematical Contest in Modeling (Top 4%) 2024
- 11th in East Division, Simon Marais Mathematics Competition 2023
- Prof Omar Wing Mem Scholarship (HK\$40,000 awarded by CUHK) 2023
- Soong Ching Ling Scholarship (¥400,000 awarded by Chinese Government) 2023
- Admission Scholarship (HK\$50,000 awarded by CUHK) 2023
- **Gold Medal, China Mathematics Olympiad (National final)** 2022
- First Prize, China Physics Olympiad (Provincial) 2022
- First Prize, China Chemistry Olympiad (Provincial) 2022

Skills

- **Languages:** Mandarin (Native), English (Fluent, IELTS 7.5 [Speaking 7]), Cantonese (Intermediate)
- **Technical Skills:** High-Performance Frameworks (vLLM, TensorRL-LLM, FSDP, Ray); Programming Languages (CUDA, Python, C++, C, Go); and Other Tools (R, SQL, LaTeX, HTML, MATLAB)